# Minimal Axioms for the Origin of Grounded Relational Structure

Checked in Lean 4

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**Status.** This document is a work in progress. It is incomplete, and all of its parts, including the axioms, definitions, results and their interpretation, are subject to revision and refinement. Passages marked in the margin were drafted with AI assistance and have not yet been checked against their sources.

*Mysteries should not be multiplied beyond necessity.*
— Eric Gans, *The Origin of Language* (1981)

## 1 Motivation and Background

### 1.1 Theories of language origin

Theories of language origin attempt to explain how a species without language came to have it. The field is broad. Among its main families are the following.

- Gestural accounts, on which language grew out of manual communication and pantomime [4, 2].
- Social accounts, on which language took over the role of grooming in maintaining group bonds [8].
- Intention-based accounts, which ground meaning in the expression and recognition of communicative intentions [13, 25].
- Game-theoretic accounts, in which signals acquire meaning as stable conventions among interacting agents [17, 23].
- Computational accounts, which locate the decisive step in the appearance of a single combinatorial operation [3].

Two problems recur across these families, and they help explain why the question of language origin has often been treated with suspicion.

**Presupposition.** Many accounts take as given, in some form, capacities closely related to the ones whose origin is at issue. This is not always an oversight; often it is a deliberate choice of what is to be explained. Three examples illustrate the point.

- *Game-theoretic accounts.* Lewis–Skyrms signaling games show how signals can acquire meaning without design or prior agreement. Skyrms extends the framework so that agents begin with no signals at all and invent new ones through a reinforcement process [23, 1]. What these models take as given is the interaction itself: a sender whose behaviour depends on a state of the world, a receiver whose action depends on the sender's behaviour, and payoffs that reward a match. The models explain which behaviour comes to mean which state. That one participant's behaviour is the kind of thing the other's action responds to *as* a signal is the frame within which they work.

- *Intention-based accounts.* These analyze meaning through the expression and recognition of communicative intentions [13]. As analyses of meaning, they are not committed to any story of origin. As a basis for one, they inherit the observation that an intention to communicate is itself a representational state, so that thought is presupposed prior to language. Tomasello's account is explicit on this point. It posits shared intentionality, joint attention and cooperative motives as a biological infrastructure that evolved before conventional language and made it possible [25]. The question of reference is moved to the evolution of that infrastructure, not answered from nothing.

- *Computational accounts.* Berwick and Chomsky locate the decisive innovation in a single combinatorial operation, Merge, and explain how hierarchical structure is built from lexical atoms [3]. They accept that words preceded grammar, and critics have pointed out that the origin of those atoms is left largely unaddressed [24].

**Unverified (AI-drafted).** The thought-priority objection to intention-based accounts needs a citation. A candidate is Devitt, "Three Mistakes About Semantic Intentions".

None of these observations is a refutation. Each account addresses its own question, and every model must take something as given, including the one developed here, which assumes perception and response. The point concerns where the line is drawn. In each case, the step by which some piece of behaviour first comes to *stand for* something is not part of what is explained.

**Evidence.** Language leaves no fossils. The events in question are prehistoric, unobservable and unrepeatable, and the indirect evidence from anatomy, archaeology, comparative cognition and genetics constrains them only weakly. In 1866 the Société de Linguistique de Paris went as far as to exclude communications on the origin of language from its proceedings [18]. Much more recently, a review by leading researchers concluded that the central questions remain largely out of reach of the available evidence [14]. Theories of origin are therefore often regarded as untestable, and dismissed as just-so stories: narratives that fit what they explain, but that no observation could refute.

These two problems point toward a particular kind of contribution. A theory of origin that cannot be tested historically may still be assessed in another way: by asking whether its conclusions *follow* from its assumptions, and which assumptions they require. A claim of the form "if these conditions hold, this structure can arise", or "must have this form", is not a claim about what happened. It can be checked deductively, and its assumptions can then be examined individually, some of them empirically, for plausibility. This does not make an origin story testable as history. It makes it accountable as an argument.

**The originary hypothesis.** Eric Gans's Generative Anthropology is notable for addressing the first problem directly, and in a way that lends itself to this kind of assessment [9, 10]. Several features distinguish it.

1. It starts from the sign itself. It asks what must happen for a sign to exist at all, not how pre-existing contents come to be communicated.

2. It derives the sign from non-semiotic behaviour. On Gans's account, the first sign is an aborted gesture of appropriation: a practical act that, by being interrupted, comes to designate instead of to seize. This gives a concrete mechanism by which representation could arise without being presupposed.

3. The sign is shared from the outset. It arises in a scene among several participants, as a mutually understood gesture, so meaning is not a private state later made public.

4. It is generative. A single minimal hypothesis is followed by a derivation of further elementary forms of language in sequence. A claimed derivation has internal structure that can be examined step by step.

*The Origin of Language* opens with the sentence "Mysteries should not be multiplied beyond necessity" [9]. It is a version of Ockham's razor, and it states the ambition of the theory: to explain the origin of the human while assuming as little as possible.

The hypothesis runs as follows. A group of hominids converge, through mimetic desire, on a single desirable object. Mutual imitation intensifies their rivalry beyond what an animal dominance hierarchy can contain, and the group faces a violent crisis. The crisis is deferred when the gesture of appropriation is aborted. The aborted gesture, shared and reciprocally understood, becomes the first sign: an *ostensive* that designates the object as a centre which no one may take. The scene concludes in the *sparagmos*, the collective division of the object.

**Unverified (AI-drafted).** The role of the dominance hierarchy in Gans's scene has not been confirmed in a source. Check Gans 1981 or 1993.

From the ostensive, Gans derives the further forms in sequence. The *imperative* arises from an ostensive made in the absence of its referent, which the hearer answers by supplying the missing object. The *declarative* arises when the imperative cannot be fulfilled, and the hearer responds with a negation of the object's presence.

**Unverified (AI-drafted).** This summary of the imperative (the "inappropriate ostensive") and the declarative (the negative ostensive) was confirmed only through secondary commentary. Check against Gans 1981.

Later work in the tradition develops related notions. Adam Katz, who also writes as Dennis Bouvard, introduced into the originary scene the *firstness* of a single participant who emits the sign before the others [16]; see also [11]. He has also developed an approach organized around the *centre* toward which the attention of a scene converges.

**Unverified (AI-drafted).** Katz on the centre has not been read directly. A candidate source is Katz, "Generative Anthropology as the One Big Discipline", *Anthropoetics* XXIV, no. 2 (2019).

## 1.2 The problem: narrative content of uncertain necessity

The present work grows out of a long engagement with the writings of Gans and Katz, and out of a specific dissatisfaction with one feature of the account. Gans's hypothesis avoids presupposing the sign. The scene that does this work, however, carries substantial narrative content:

- *convergence*: mimetic desire draws the participants to a single object;
- *crisis*: mimetic rivalry exceeds what an animal dominance hierarchy can contain;

- *deferral*: the prospect of violence motivates the abortion of the gesture of appropriation;
- *centre*: the designated object becomes a sacred centre;
- *resolution*: the scene ends in the sparagmos, in which the tension the sign has deferred is discharged on the object, not on other members of the group.

Much of this content is inherited from René Girard's mimetic theory [12]. Gans accepts that language originates in a mimetic crisis, while rejecting the scapegoat mechanism as sufficient to produce a sign. The sparagmos is in part his way of incorporating Girard's account of sacrificial violence. Each element is anthropologically suggestive, and each ties the origin of language to themes the theory develops elsewhere, such as the sacred, ritual, ethics and exchange.

Which of these elements is actually needed is a natural question for an account that begins by refusing to multiply mysteries. It has been raised within the tradition itself. Sackeroff, for example, identifies the mechanism of deferral as a point of difficulty, observing that the account draws on two different sources of motivation: animal appetite for the object, and a distinctively human investment in the sign [22]. The present work takes up this internal question in a formal setting.

Several of the scene's elements are open to objection on their own terms.

- *Violence.* Among primates, conflict is frequent but usually limited, and it is often followed by reconciliation [6]. Evolutionary game theory explains why contests between members of the same species usually take the form of limited, not escalated, fighting [19]. Disputes that end in exhaustion, hierarchy or reconciliation do not by themselves produce signs. The account's answer is that mimetic intensity has disabled these ordinary resolutions, which is itself one of the elements whose necessity is in question.

- *Deferral as a strategy.* In the Hawk–Dove game, mutual restraint is not stable: against a participant who refrains, taking the resource does better. A deferral that rests on self-interest therefore needs further explanation. The account supplies it through the intensity of mimetic desire and the threat of collective violence, again through the narrative content.

- *Division.* By the time the sparagmos begins, the aborted gesture has already designated the object. Its role in the origin of distribution and of the sacred is not in question here. Its role in the emergence of reference is unclear.

The issue is therefore not whether these elements are plausible. It is whether they are *load-bearing*. If the derivation of the sign and its forms depends on them, it inherits their contingency, and the charge of a just-so story applies. If it does not, the derivation should be statable without them.

The present work follows in the spirit of Gans's opening sentence, and applies it to the originary scene itself. Mimetic rivalry, catastrophic violence and the sparagmos are explanatory commitments, and each is a candidate for a mystery multiplied beyond necessity. The question is whether the derivation of the sign and its forms requires them. If it does not, the principle with which Gans begins recommends setting them aside.

**Unverified (AI-drafted).** This description of the sparagmos comes from an *Anthropoetics* page whose author has not been confirmed. Find a citable source.

**Unverified (AI-drafted).** The relation between Gans and Girard summarized here has been confirmed only through an encyclopedia entry. Find a primary or scholarly source.

**Unverified (AI-drafted).** The issue number of Sackeroff's article was inferred from its URL. Confirm.

**Unverified (AI-drafted).** The Hawk–Dove game postdates Maynard Smith and Price (1973). Cite the right source, probably Maynard Smith, *Evolution and the Theory of Games* (1982).

### 1.3 The target: a grounded symbolic graph

A generative account needs a destination as well as a starting point. Both can be stated using accepted notions of the sign.

**The starting state.** There are actors and physical objects. There are contexts: at each time, which actors are present and which objects are within whose reach. Actors move, and some movements are directed at objects. All the relations in this state are momentary and local. Each holds between particular actors and things, at a particular time, in a particular context. There are no symbolic nodes.

**The sign as a relation.** The accepted definitions of the sign are relational. For Saussure, the linguistic sign joins a concept and an acoustic image, and its value is fixed only by its differences from other signs in the system. For Peirce, "a sign, or representamen, is something which stands to somebody for something in some respect or capacity" [21, 2.228]. This is a relation among a sign, an object and an interpretant, and it holds *for* someone. Morris divides the study of signs by the relations a sign can enter into: to other signs, to objects, and to its interpreters [20].

**Unverified (AI-drafted).** Add a citation, with page, to the *Course in General Linguistics*.

On each account, a sign is not a thing with a property but a term in relations. Among those relations are relations to the *actors* who produce and interpret it. The minimal mathematical structure of relations is a graph. Relations among more than two terms can be represented as a hypergraph, or encoded in an ordinary graph by an incidence construction, in which each instance of a relation becomes a node joined to its arguments. The relevant graphs are many-sorted: their vertices include actors, physical objects, and the acts by which actors relate to objects and to one another.

**Actors, time and context.** Every such relation is indexed by actors, by a time and by a context. The most elementary sign relation, ostension, shows all three at once. *This*, *here* and *now* designate only within the situation in which they are produced, for those who share it. A pointing gesture relates one actor, by way of a movement, to an object, at one moment and in one place, and only for another actor who takes it up. The relation does not outlast the act.

Symbolic reference is what escapes this indexing. Hockett lists *displacement*, the capacity to refer to what is removed in time or place from the moment of speaking, among the design features of human language [15]. Deacon, building on Peirce, argues that symbolic reference is grounded in indexical reference but depends on relations among symbols themselves [7]. In graph terms, a symbolic node is something that persists across contexts and across occasions of use: different actors can re-identify it, at other times and in other places, and it can enter into further relations there.

**The end state.** A generative account must arrive at a graph with the following properties.

1. *Nodes that stand for things.* Some vertices are symbolic, and stand for objects to actors.

2. *Persistence beyond the context of origin.* A symbolic node can be re-identified at another time, in another place and by another actor, including in the absence of what it stands for.

3. *Relations among symbolic nodes.* Symbolic nodes stand in relations to one another, not only to objects, and some of these relations have *roles*, where it matters which term is which.

4. *Grounding.* Every symbolic node can be traced back to the physical world, so there are no free-floating symbols.

5. *Truth and falsity.* What is said with symbolic nodes has conditions under which it holds, and it can fail.

6. *Partiality.* Relations can be affirmed, denied, or left unsaid, without silence having to carry information.

7. *Open-endedness.* Any finite relational structure over grounded objects can be expressed.

**Unverified (AI-drafted).** Hockett and Altmann's "prevarication" may be the right citation here. Check.

**The generative task.** The task is to get from the starting state to the end state using only what the starting state contains, and operations on it applied to their own products. Nothing symbolic may be assumed along the way. Several features commonly attributed to human language lie outside the target: the vocal-auditory channel, duality of patterning, and cultural transmission across generations. The account addresses the emergence of reference and of relational structure, not of speech.

The target also makes precise a hypothesis that motivated the work:

- the ostensive establishes an edge between actors and an object within a single context;
- the imperative is such an edge whose object is missing from the actors' current context;
- the declarative yields a node: a relation that has itself become something the actors can refer to, and that persists across contexts.

The formalization supports a sharper version of the last point. The first symbolic node is a *reified relation*, and relations between nodes arise by the same operation applied one level up.

### 1.4 Aim: a minimal axiom set that does not license itself

The aim is to find a minimal set of axioms under which the end state of Section 1.3 can be reached from the starting state, and to prove that it can.

**The rule against self-licensing.** An axiom set *licenses itself* when what it is supposed to generate is built into its assumptions, so that the derivation only returns what was put in. To exclude this, every axiom set considered here must satisfy two conditions.

1. *Base vocabulary only.* Axioms may speak only of what the starting state contains: actors, physical objects, times, movements, reach, contact, perception,

and resemblance between movements. They may not mention signs, meaning, reference, uptake, the ostensive, the imperative, the declarative, symbolic nodes, claims or truth, nor any notion defined from these.

2. *Target notions by definition only.* Every notion that belongs to the end state is introduced by an explicit definition in the base vocabulary. A definition adds no constraint on what can happen: it names a pattern, which may or may not occur.

Under these conditions, every property of the end state that the account establishes is a theorem, derived from axioms that do not mention it.

**Definitions are hypotheses.** The rule shifts the weight of the account onto its definitions: what it is for a movement to be taken up as being about something, or for a pattern of responses to constitute a shared centre. A definition can smuggle in as much as an axiom can. Two safeguards apply. Definitions are drawn as closely as possible from accepted notions of the sign, and they are stated as explicit hypotheses about what the defined notion consists in, open to dispute and to replacement by alternatives.

**Generativity relative to the target.** With the target fixed and the rule in force, the aim has three parts.

**Existence.**
There is a model of the axioms in which the end state, or the relevant part of it, arises from a starting state that lacks it.

**Necessity.**
In every model of the axioms, whenever part of the end state exists, it has a determinate origin and order.

**Universality.**
The operations that produce the first symbolic structures suffice for open-endedness: any finite relational structure over grounded objects can be reached.

**Minimality.** An axiom set is minimal when each axiom is needed: for each axiom there is a model of the others in which some property of the end state fails. Minimality is shown axiom by axiom, by exhibiting such countermodels.

The aim is not to show that the end state was inevitable, nor that it was reached in one particular way. It is to show that it can be reached under assumptions that are weak, explicit and free of the target, and to identify which assumptions each part of it requires.

**Comparing accounts.** The same rule suggests a way of comparing accounts. Formalize an account, separate its primitives into base vocabulary and anything semiotic, and ask whether its target notions can be defined from the base part alone. Padoa's method from logic gives a standard test: a notion is not definable from the other primitives if two models agree on all of them but disagree on that notion. One feature that may signal self-licensing is a primitive kind of act whose

only role is to be responded to, such as the costless signals of a signaling game. Any such result holds only relative to a particular formalization, and only relative to the present target, the emergence of sign-hood. It is not a criticism of what other accounts set out to explain. Applying the test to other accounts, including Gans's own scene, is left as work in progress.

### 1.5 Rival orderings

A further motivation is comparative. The sequence ostensive, imperative, declarative is one reading among several, and other readings, for instance ones in which the declarative is primary, are possible. Under the rule of Section 1.4, the choice between them becomes precise. An account that builds a particular mode into its axioms licenses its own ordering. The real question is which orderings can be generated from axioms in the base vocabulary, and under which assumptions. The framework developed here bears on this directly. In it, the declarative requires the *absence* of its object, but not a prior failed imperative. The imperative and the declarative appear as two responses to absence, not as successive stages.

**Unverified (AI-drafted).** Name proponents of declarative-first or other orderings, or rephrase as the author's conjecture.

### 1.6 Why machine verification

Arguments in humanistic theory often rely on steps that are intuitively compelling but unstated. Formalization in a proof assistant forces every assumption to be written down and every inference to be checked. All results reported here are formalized in the Lean 4 proof assistant [5] and checked in full, with no unproven steps.

The process was itself informative. Several assumptions that the informal narrative leaves implicit were exposed only when a proof failed without them. Among them:

- a kinematic principle linking a reproduced gesture to the object it was originally directed at;
- the requirement that one participant act before another, which corresponds to Katz's firstness;
- the dependence of falsehood on the drift of signs away from their iconic origins;
- the reliance of affirmation, derived as the negation of a negation, on classical logic.

Formalization also makes it possible to say exactly which statements are derived and which are merely definitions. The paper marks this distinction throughout.

### 1.7 Outline

The remaining sections are less polished than this one. Section 2 summarizes the formal system: its primitives, axioms and definitions. Section 3 lists the results, with the names of the corresponding theorems in the Lean development. Section 4 records how the account developed, including the main decisions and the assumptions that were abandoned along the way. Section 5 collects findings of particular interest, especially concerning firstness and Gans's account. Section 6 lists open problems. The Lean development is in the `lean/` directory of the repository. Table 1 shows where each property of the end state is reached.

| Property of the end state | Where it is reached |
|---|---|
| 1. Nodes that stand for things | declarative and lineage (Sections 3.1–3.2) |
| 2. Persistence | re-presentation; displaced claims (Sections 3.2, 3.4) |
| 3. Relations, with roles | claims and ordered claims (Sections 3.4–3.5) |
| 4. Grounding | content traces to an object once within reach (Section 3.1) |
| 5. Truth and falsity | falsehood requires drift (Section 3.3) |
| 6. Partiality | polarity; partial universality (Section 3.6) |
| 7. Open-endedness | universality (Section 3.4) |

Table 1: Where each property of the end state is reached.

## 2 The Formal System

This section summarizes the current system, formalized in `lean/GaEmergence/Completion.lean` and the files built on it.

**Worlds.** A world is a single discrete-time history over a type $P$ of actors, a type $O$ of physical objects and a type $M$ of movement forms. Its primitives are:

- $\mathsf{mv}(t, p)$: the form of $p$'s movement at time $t$, if $p$ moves;
- $\mathsf{heads}(t, p, o)$: $p$'s movement at $t$, if continued, would end in contact with $o$;
- $\mathsf{contact}(t, p, o)$ and $\mathsf{within}(t, p, o)$: physical contact, and reach;
- $\mathsf{perceives}(t, q, p)$: $q$ perceives $p$'s movement at $t$;
- $\mathsf{opens}(m', m)$ and $\mathsf{matches}(m', m)$: form $m'$ begins like $m$, and $m'$ fully reproduces $m$;
- a response window $w$, and an unspecified retention predicate.

Presence is reach: an object is present to an actor when it is within that actor's reach.

**Axioms.** All axioms are stated in the base vocabulary.

A1. A movement can head only toward what is within reach.
A2. Only a movement can head toward anything.
A3. Contact completes a heading.
A4. Perception is first-order: an actor perceives another actor's actual movement.
A5. matches is reflexive and symmetric, but not assumed transitive; a full match also begins like its model.
A6. Retention, left unspecified: a retained movement recurs in a matching form.
A7. Kinematics: a movement that begins like a reach for $o$, made where $o$ is within reach, is itself a reach for $o$.

Variation is implicit: no axiom requires or forbids any particular movement.

**Definitions.** All target notions are definitions.

- *Directedness* is a pending completion. A movement is directed at an object if it heads for it, and at an earlier movement if it begins like it. *Completion* is contact at the next step, or a full match. A movement is *suspended* toward a target if it is directed at it and does not complete.

- *Response*: $q$'s movement at $t'$ responds to $p$'s movement at $t$ if $q$ perceived it and $t < t' \leq t + w$.
- *Taking-as* (the response principle): $q$ takes $p$'s movement as being about $\tau$ if $q$ responds with a movement directed at $\tau$.
- *Centre*, with firstness: $p$ is suspended toward a target, $q$ perceives this and is suspended toward the same target strictly later within the window, and $p$ perceives that.
- *Dependent response*: regularity (another such episode, with the same kind of movement by $p$) and contrast (an occasion where $q$ could have responded, $p$ made no such movement, and $q$ did not respond).
- *Ostensive*: a centre on an object, with a dependent response. *Declarative*: a centre on a movement, with a dependent response. *Imperative*: a movement taken as being about an object out of the mover's reach.
- *Lineage*: a movement stands for an object if it is the first movement of an ostensive on it, or fully matches an earlier movement that stands for it.
- *Graph notions.* A *symbolic node* is a movement that is the target of a declarative. An *arc* runs from a node to the object it stands for.
- *Claims and their truth.* A *claim* is a centre on two targets at once, each with a dependent response. An *ordered claim* is a centre on one target, followed by a centre on another. *Together*: someone has both objects within reach at the same time.
- *Saying and assertion.* Saying is recursive: a claim says "not together", and a centre on a movement that says $\varphi$ says "not $\varphi$". A saying is *asserted* if no later centre targets a movement of the same kind.

# 3 Results

All results below are checked in Lean, with no unproven steps. The labels mark each result as *definitional* (a direct consequence of the definitions), *derived* (it uses the axioms), or a *model* (an existence result or countermodel, shown by constructing a world). Theorem names are given in typewriter type.

## 3.1 Meaning and origin

- A centre is exactly a taking-as whose response is suspended, so the centre is not a separate notion: `joint_iff_suspended_taking`, `center_takes`. *Definitional.*
- A simultaneous joint stop is not a centre, because nobody responds: `simultaneous_not_center`. A one-off coincidence is not dependent: `coincidence_not_dependent`. Nor is a response from someone who holds back whenever possible: `indiscriminate_not_dependent`. *Definitional.*

- Completion ends aboutness: `completion_ends_aboutness`. *Definitional; formerly an axiom.*
- Whatever a sign stands for was once within someone's reach: `content_was_present`. *Derived, by induction.*
- Without recurrence there are no symbolic nodes: `no_recurrence_no_symbol`. *Derived.*

### 3.2 Imperative and declarative

- A movement made where its object is out of reach is not directed at it: `aimless_in_absence`. *Derived.*
- Taking a movement as being about an object requires a response where the object is, so fetching is derived, not assumed: `takes_object_within`. When nothing is in reach, only movements can be targets: `unreachable_only_acts`. *Derived.*
- *Salience.* Pure attention to a gesture descended from a reach requires the object's absence: `salience`. *Derived, using kinematics.*
- An absent object can be addressed only through a sign: `absent_only_through_sign`. *Derived.*
- *Direction of fit.* The imperative is unsatisfied when made, and satisfying it requires the world to change: `imperative_needs_change`. A pure declarative's "don't be present" is satisfied when made: `declarative_satisfied_when_made`. *Derived.*
- *The interdiction redirected.* In the ostensive the object is within reach and the participants refrain. In the declarative the object is out of reach: `interdiction_redirected`. *Derived.*

### 3.3 Truth and falsehood

- With kinematics, and if onset-matching survives full matching (coherence), no declarative is false: `no_falsehood_without_drift`. *Derived.*
- In a false declarative, the movement made in the object's presence does not begin like any earlier reach for it: `false_requires_arbitrary`. *Derived.*
- A drifting lineage yields a false declarative in a world satisfying every axiom: `falsehood_from_drift`. *Model.*

### 3.4 Graphs

- A claim on two movements is exactly two declaratives made by the same movements: `claim_iff_two_declaratives`. *Definitional.*
- Claims are grounded: `claim_grounded`. *Derived.*

- *Universality.* For any $n$ objects, any loop-free graph $G$ on them and any arrangement $H$ of which pairs are together, there is a history satisfying every axiom in which every object has a grounded sign, every edge of $G$ is a claim with arcs to exactly its endpoints' signs, each claim is true iff its objects are not together in $H$, and there are no other claims: `universality`. Claims are made where neither speaker has any object in reach (`Univ.displaced`). *Model, for every $G$ and $H$.*

### 3.5 Roles

- An unordered claim is the same event read either way round: `claim_symm`. Hence every faithful reading of unordered claims expresses a symmetric relation: `unordered_only_symmetric`. *Definitional and derived.*
- Succession, defined from reach and time, is asymmetric. In a concrete world it has no faithful unordered reading: `succession_not_unordered`. *Model.*
- Ordered claims that determine what they are about give every relation a faithful reading: `ordered_faithful`. In a model, "A then B" is true and "B then A" is false: `succession_ordered`, `Ord.order_matters`. *Derived and model.*

### 3.6 Polarity

- Everything said is a negation: `says_neg`. A centre on a claim says "not (not together)", which holds exactly when the objects are together: `affirm`. This step uses classical logic. A formula and its negation are never both true: `no_contradiction`. *Definitional and derived.*
- *Partial universality.* For any pairs to deny, pairs to affirm and any arrangement, there is a history in which exactly those pairs are asserted, each true iff the arrangement agrees, and all other pairs are unsaid: `partial_universality`. *Model, for every choice.*

### 3.7 Models and countermodels

- `emergence`: an ostensive (two episodes and a contrast), a satisfied imperative, and a pure declarative standing for an absent object, with non-transitive matching.
- `gaze_not_center`: joint looking, with the object in reach and mutual perception, is not an ostensive.
- `kinematics_needed`, `fit_needs_kinematics`: without kinematics, a "declarative" can occur with the object present, and its "don't be present" fails.

## 4 How the Account Developed

The account went through several versions. Each version is preserved in the Lean development. The main decisions and changes, in order:

1. *The starting axioms.* The work began with four informal axioms: plurality, contextual recognition, contextual recurrence, and a forced preservation of failed relations. Examining them showed the following.
   - Plurality is derivable once recognition is required to be by someone else.
   - Recognition needs a target, and a ground; otherwise meaning could arise from nothing.
   - Recurrence must not be an equivalence relation, or sign-types are smuggled in, and it must carry its target over.
   - Forced preservation splits into a compulsion to attempt and the untenability of indefinite failure. If forcing never lapses, these together entail resolution, so "success is not guaranteed" requires that forcing can lapse.
   - A reflexive uptake, attention to a sign as such, was needed. Without it there are no symbolic nodes, and forcing on an unavailable object becomes inconsistent.

   (`Basic.lean`, `Theorems.lean`, `Models.lean`.)

2. *The ostensive was being assumed.* Presence was a primitive ground for recognition. It was replaced by a practical layer of reaching, grasping, mimetic symmetry and restraint. The ostensive became a joint deferral, entailed by convergence, symmetry and restraint. Countermodels with hierarchy and with rivalry showed that each ingredient was needed. (`Originary.lean`.)

3. *Uptake was a primitive record.* It was replaced by a definition: taking-as as a kind of response. (`Behaviour.lean`.)

4. *Catastrophe was questioned.* Violence is common and is often resolved by fatigue. Restraint was replaced by exhaustion: clashes cannot recur indefinitely. The ostensive then follows when desire outlasts the capacity to fight. Countermodels covered endless fighting, walking away, and hierarchy. (`Behaviour.lean`.)

5. *Refraining as a losing strategy.* On ordinary assumptions about preference, mutual restraint is unstable. A division of the object afterwards (the sparagmos) was considered as the payoff and rejected as an unnecessary narrative commitment. Chance, meaning that any movement may occur, was adopted instead: a joint holding-back can simply happen.

6. *What makes a sign worth repeating?* Retention was introduced as an unspecified function, mirroring the unspecified forcing. Without repetition, only ostensives can exist. (`Retention.lean`.)

7. *Why reaching and holding back?* Appropriation and rivalry were replaced by completion and the principle that completion ends aboutness. Gaze was excluded by defining directedness as a pending contact. (`Center.lean`.)

8. *A self-licensing axiom was removed.* An axiom compelling a fetch when a sign recurred in its object's absence built the imperative into the axioms. It was replaced by the response principle, from which fetching follows. Forcing was then dropped altogether: the permission to continue a failed relation needs no axiom, because variation already provides it.

9. *The completion system.* Directedness became a pending completion, either contact or match. Kinematics was found to be necessary when the proof of salience failed without it. (`Completion.lean`.)

10. *The imperative is not a prerequisite.* The declarative requires the absence of its object, not a failed imperative. Gans's "don't", addressed to the object, led to the direction-of-fit results. (`Fit.lean`.)

11. *Falsehood was at first impossible.* Every pure declarative came out true. Drift, and the arbitrariness it produces, turned out to be the precondition of falsehood. (`Falsehood.lean`.)

12. *From nodes to graphs.* A claim became a node with outgoing "about" arcs, not an arc between nodes. This keeps a single kind of arc, accommodates symmetric relations and relations with more than two terms, and leads to universality. (`Graphs.lean`.)

13. *Firstness.* Once it was noticed that a simultaneous centre contains no uptake, a centre was required to have one participant first and another responding, following Katz. Lineage became rooted in the first movement.

14. *Dependent response.* "In response to" was strengthened from "perceived, then a movement within the window" to regularity with contrast. A counterfactual criterion was rejected because it would need dispositions as primitives. A single-episode mark was rejected because coincidences leave the same marks.

15. *Roles and polarity.* Ordered claims and succession gave relations with roles (`Roles.lean`). The closed-world problem of negative-only claims led to the positive as a negation of a negation, and to partial universality (`Polarity.lean`).

16. *The paper.* The rule against self-licensing, the target end state, and the convention for marking unverified passages were made explicit while writing Section 1.

# 5 Findings

## 5.1 Firstness

1. *Firstness is required by the notion of response.* A simultaneous joint holding-back contains no uptake, because neither actor responds to the other. For a centre to contain any taking-as, one actor must act first and the other respond. Katz's firstness does not need to be posited. It follows from what a response is.

2. *Firstness reappears as order inside claims.* Unordered claims can express only symmetric relations. Temporal order inside a claim, which is firstness at the level of the claim, is what makes relations with roles expressible.

3. *The first actor fixes the form.* The first actor's movement is the sign and the second's is the uptake, so lineage descends from the first. This gives a

minimal, non-political reading of the importance Katz assigns to firstness. The stronger reading, on which the first holds authority, is not needed by the derivation.

4. *Firstness locates the risk.* Holding back first is exposed, since the other may simply take. Only the response turns the first holding-back into a sign.

### 5.2 Gans's account

1. *The aborted gesture survives as the core.* When shared reference to a centre is required to be peaceful and the same for both actors, joint holding-back is what remains, and the ostensive no longer needs to be assumed.

2. *Holding back is needed for a different reason.* It is needed not because of violence, but because completion ends aboutness: a completed reach is getting, not pointing. Desire and rivalry become one possible motive for holding back, not a condition of it.

3. *The narrative content is not load-bearing for these results.* Mimetic crisis, catastrophic violence, mimetic symmetry and the sparagmos are not used by the possibility, necessity or universality results. Chance and retention take their place. In the spirit of Gans's opening sentence, they are mysteries that need not be multiplied.

4. *Gaze is not enough.* Joint looking at a reachable object, with mutual perception, is not an ostensive. There is no completion to withhold. This supports Gans's choice of a gesture of appropriation over attention alone.

5. *The imperative is a sibling of the declarative, not its precursor.* The declarative requires the object's absence, not a failed imperative. Gans's own description, in which the "don't" is addressed to the object, fits this reading well.

6. *The declarative is an imperative already obeyed.* This is now a theorem about direction of fit. The imperative is unsatisfied when made and fits the world only after it changes. The declarative's "don't be present" is satisfied when made.

7. *The interdiction is redirected.* In the ostensive, the actors refrain from the present object. In the declarative, the object "refrains" from being present.

8. *The declarative needs a kinematic principle.* Reproducing a reach where the object is reachable reaches it. Without this principle, attention to a gesture as such could occur with the object present, and the tie between the declarative and absence is lost.

9. *The first signs are always true.* Falsehood requires drift away from the shape of the reach the sign descends from. Arbitrariness is acquired, not original, and it is the precondition of falsehood.

10. *Necessity becomes "the only possible form".* Gans presents the declarative as what must happen when the imperative fails. Here, nothing compels anyone to continue. But if a failed relation is continued, the response can only take

the form of the imperative (if the object can be reached) or of the declarative (if it cannot).

11. *A sign needs repetition.* A one-off coincidence is not a sign. The first holding-back becomes a sign retroactively, when it is repeated and answered.

12. *The node is a reified relation.* The first symbolic node is a relation made into something one can refer to. A claim is the same operation one level up, and it is nothing more than two declaratives made by the same movements.

13. *Negation comes first, and affirmation costs classical logic.* The priority of the "don't" is formally coherent. Everything said is a negation, and the positive is a negation of a negation. Recovering the positive from it uses excluded middle.

14. *Many natural axioms secretly mention uptake.* Several candidate axioms, including forced preservation and reflexive uptake, turned out to presuppose the sign. This motivated the rule against self-licensing.

# 6 Open Problems

- *Minimality.* Countermodels are still needed for the physical axioms, for perception and for the resemblance axioms.

- *Likelihood.* The results are about possibility. A probabilistic or dynamic version, with retention and variation given quantitative form, is open.

- *Response.* Regularity with contrast is a minimal criterion: two episodes and one contrast, and common causes are not excluded.

- *Two kinds of completion.* Contact and match are only partly unified, and "incomplete match" needs a more precise account.

- *Scope.* The account covers two actors at a time, coarse time, and co-presence realized by passive "witnesses". Roles exist only for succession, and polarity only for togetherness. Properties, quantification and deeper recursion of claims are open.

- *Comparing accounts.* Apply the self-licensing test to other accounts, including Gans's own scene and Lewis–Skyrms signaling games. It is conjectured that the present account generates the frame that signaling games presuppose.

- *Verification.* The open items in `VERIFYING.md` need checking, along with the interpretations of Gans and Katz against their texts.

- *Housekeeping.* The earlier layers of the Lean development use different definitions, and could be archived or unified with the current system.

## References


[1] J. M. Alexander, B. Skyrms and S. L. Zabell. Inventing new signals. *Dynamic Games and Applications*, 2012. doi:10.1007/s13235-011-0027-2.

[2] M. A. Arbib. *How the Brain Got Language: The Mirror System Hypothesis*. Oxford University Press, New York, 2012.

[3] R. C. Berwick and N. Chomsky. *Why Only Us: Language and Evolution*. MIT Press, Cambridge, MA, 2016.

[4] M. C. Corballis. *From Hand to Mouth: The Origins of Language*. Princeton University Press, Princeton, 2002.

[5] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, pages 625–635. Springer, 2021.

[6] F. de Waal. *Peacemaking among Primates*. Harvard University Press, Cambridge, MA, 1989.

[7] T. W. Deacon. *The Symbolic Species: The Co-evolution of Language and the Brain*. W. W. Norton, New York, 1997.

[8] R. Dunbar. *Grooming, Gossip, and the Evolution of Language*. Harvard University Press, Cambridge, MA, 1996.

[9] E. Gans. *The Origin of Language: A Formal Theory of Representation*. University of California Press, Berkeley, 1981. New edition: *The Origin of Language: A New Edition*, with an introduction by A. Katz, Spuyten Duyvil, New York, 2019.

[10] E. Gans. *Originary Thinking: Elements of Generative Anthropology*. Stanford University Press, Stanford, 1993.

[11] E. Gans. On firstness. In A. Katz, editor, *The Originary Hypothesis: A Minimal Proposal for Humanistic Inquiry*, pages 45–57. The Davies Group, Aurora, CO, 2007.

[12] R. Girard. *Violence and the Sacred*. Translated by P. Gregory. Johns Hopkins University Press, Baltimore, 1977.

[13] H. P. Grice. Meaning. *The Philosophical Review*, 66(3):377–388, 1957.

[14] M. D. Hauser, C. Yang, R. C. Berwick, I. Tattersall, M. J. Ryan, J. Watumull, N. Chomsky and R. C. Lewontin. The mystery of language evolution. *Frontiers in Psychology*, 5:401, 2014.

[15] C. F. Hockett. The origin of speech. *Scientific American*, September 1960.

[16] A. Katz. Remembering Amalek: 9/11 and generative thinking. *Anthropoetics*, X(2), Fall 2004/Winter 2005.

[17] D. Lewis. *Convention: A Philosophical Study*. Harvard University Press, Cambridge, MA, 1969.

[18] J. Lyons. The origin of language, speech and languages. In *Natural Language and Universal Grammar: Essays in Linguistic Theory*, volume 1. Cambridge University Press, Cambridge, 1991.

[19] J. Maynard Smith and G. R. Price. The logic of animal conflict. *Nature*, 246:15–18, 1973.

[20] C. W. Morris. *Foundations of the Theory of Signs*. International Encyclopedia of Unified Science. University of Chicago Press, Chicago, 1938.

[21] C. S. Peirce. *Collected Papers of Charles Sanders Peirce*. Harvard University Press, Cambridge, MA. Cited by volume and paragraph.

[22] S. Sackeroff. The ends of deferral. *Anthropoetics*, XX(2), 2015.

[23] B. Skyrms. *Signals: Evolution, Learning, and Information*. Oxford University Press, Oxford, 2010.

[24] M. Studdert-Kennedy and H. Terrace. In the beginning: a review of Robert C. Berwick and Noam Chomsky's *Why Only Us*. *Journal of Language Evolution*, 2(2):114–125, 2017.

[25] M. Tomasello. *Origins of Human Communication*. MIT Press, Cambridge, MA, 2008.